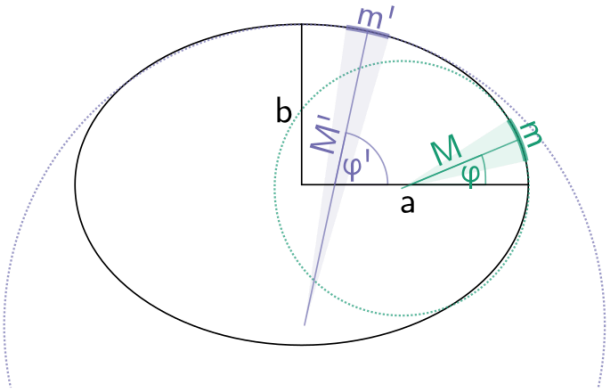
L4: Earth Ellipsoid, ellipsoidal coodinates, and satellite coordinate systems

Historic Arc measurement



meridional radius of curvature

semimajor axis

$$M = \frac{a(1-e^2)}{\left(1-e^2\sin^2(\varphi)\right)^{\frac{3}{2}}}$$

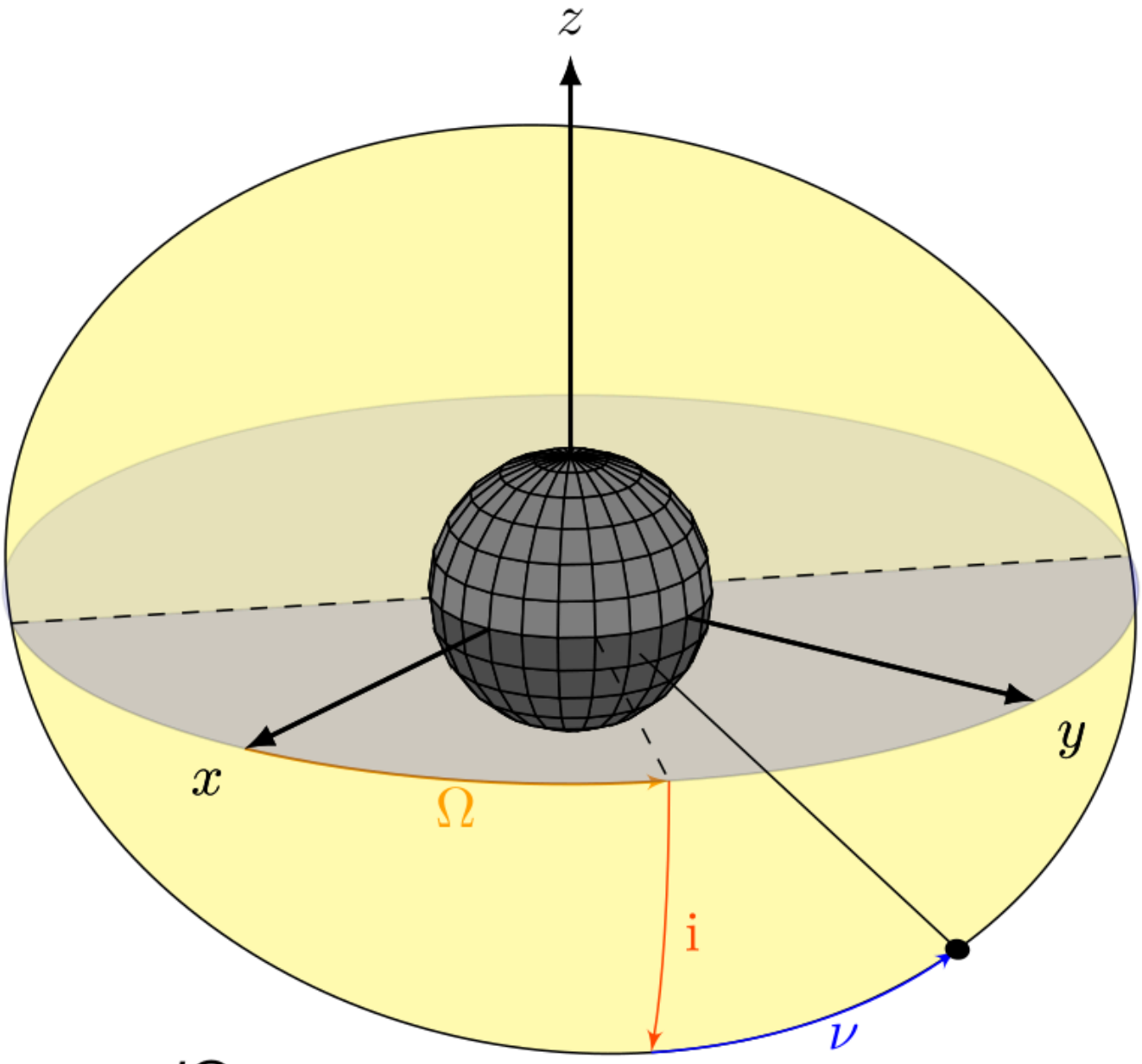
1st eccentricity

ellipsoidal latitude

Location	Latitude	Arc length (toises)
Quito	0°0′	56.751
Cape of Good Hope	33°18′	57.037
Rome	42°59′	56.979
Paris	49°23′	57.074
Lapland	66°19′	57.422

ROYAL ASTRON. SOC./SPL, Robinson 2011

Precession of Orbital Plane



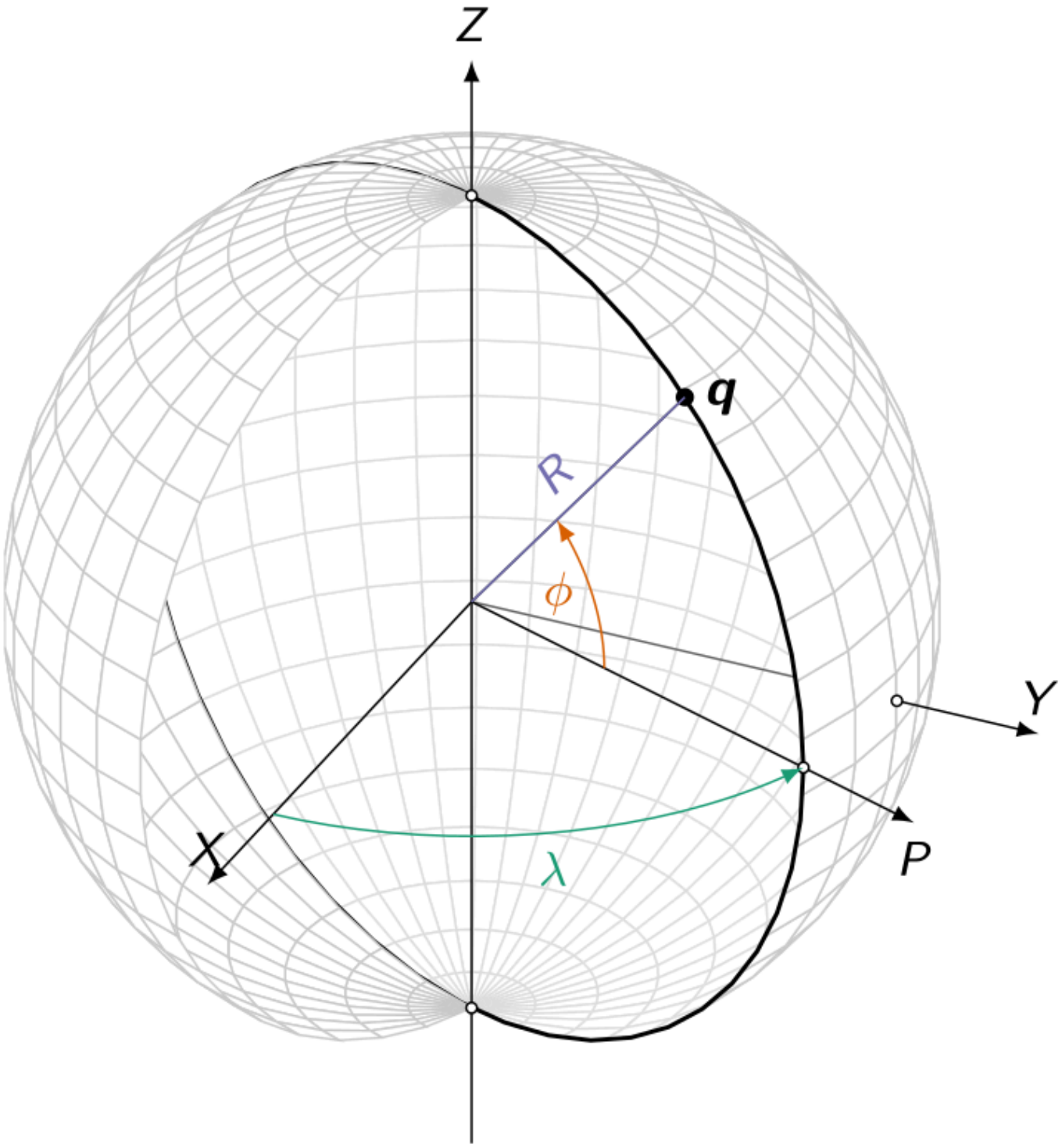
$$\dot{\Omega} = \frac{d\Omega}{dt} \neq 0$$

- From orbital theory: rotation rate of $\frac{d\Omega}{dt} = 3.2^\circ d^{-1}$ predicted
  - Confirmation by Sputnik 1 (1957)
- $\exists$ Historic measurements of Earth oblateness

Requirements

- Reference surfaces in geodesy require a definition precise at mm-level

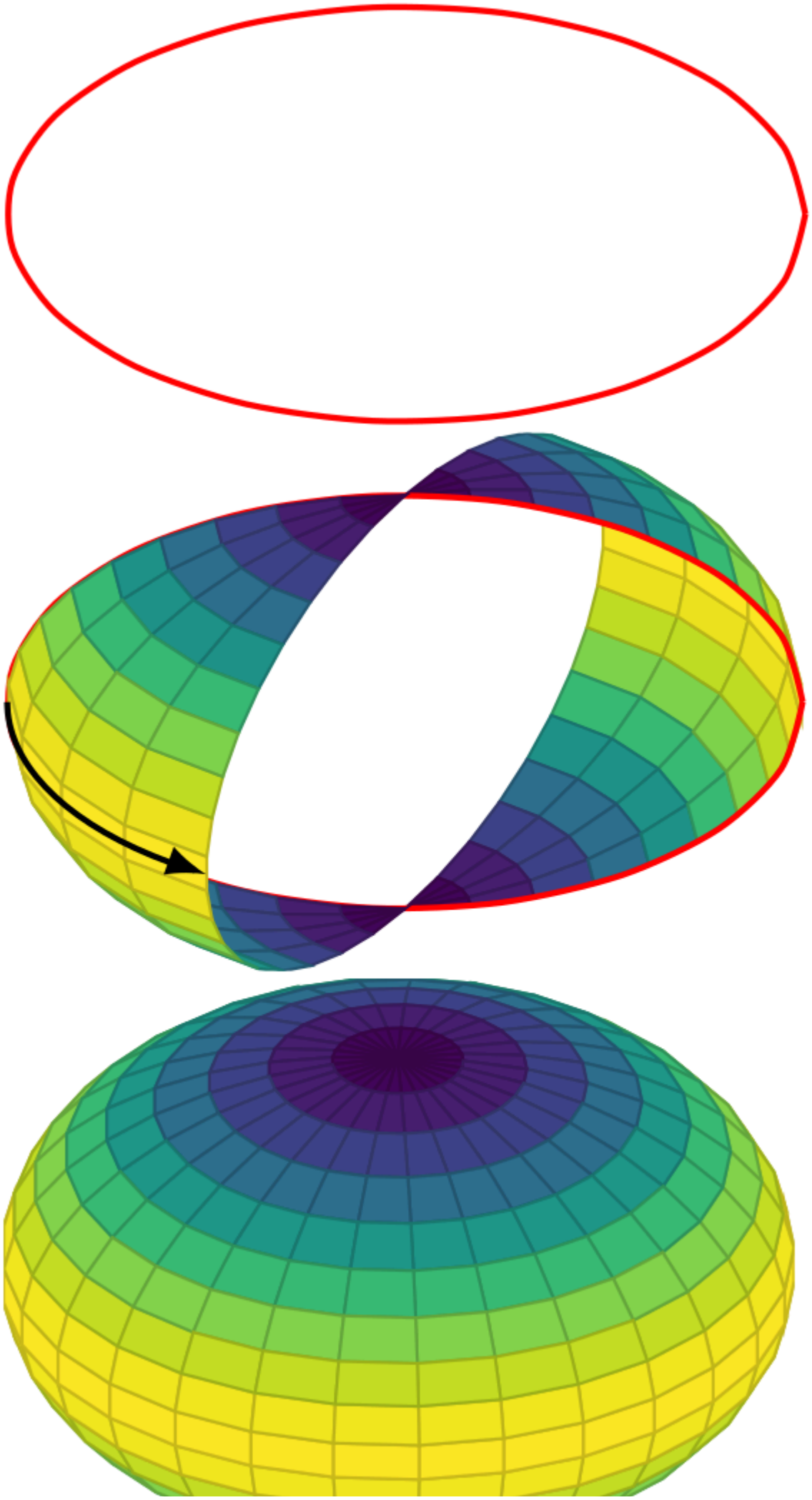
Ellipsoidal coordinates



Points on Sphere

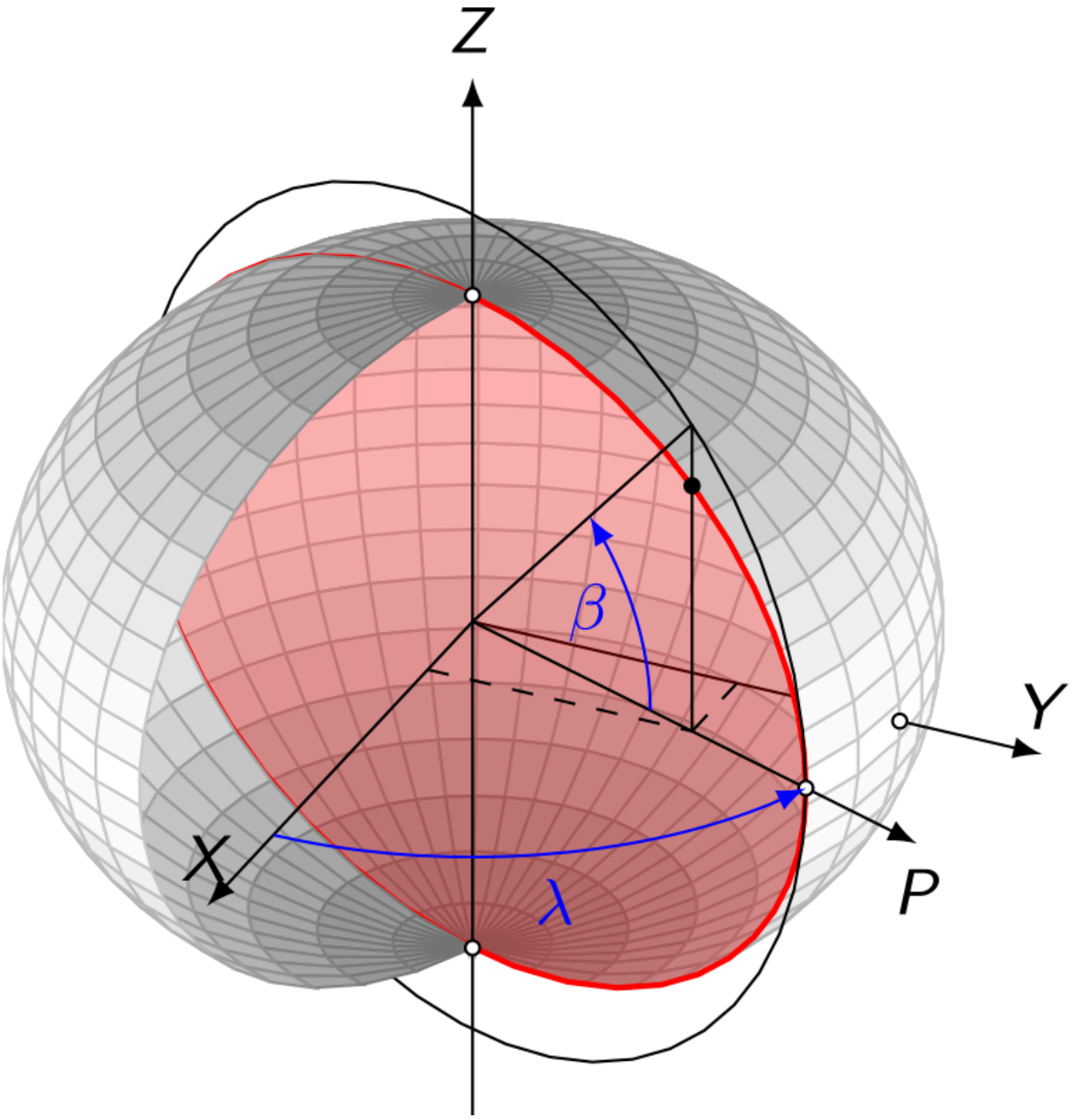
$$\boldsymbol{q}(\lambda, \phi) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} R\cos\lambda\cos\phi \\ R\sin\lambda\cos\phi \\ R\sin\phi \end{bmatrix}$$
$$\lambda \in [0^\circ, 360^\circ)$$
$$\phi \in [-90^\circ, 90^\circ]$$

- For possible parametric representation (RHS)
- ϕ becomes β
- β parametric/reduced latitude, hard to interpret goemetrically



P-Axis

- Exploit that ellipsoid of revolution is rotational symmetric around z-axis by construction

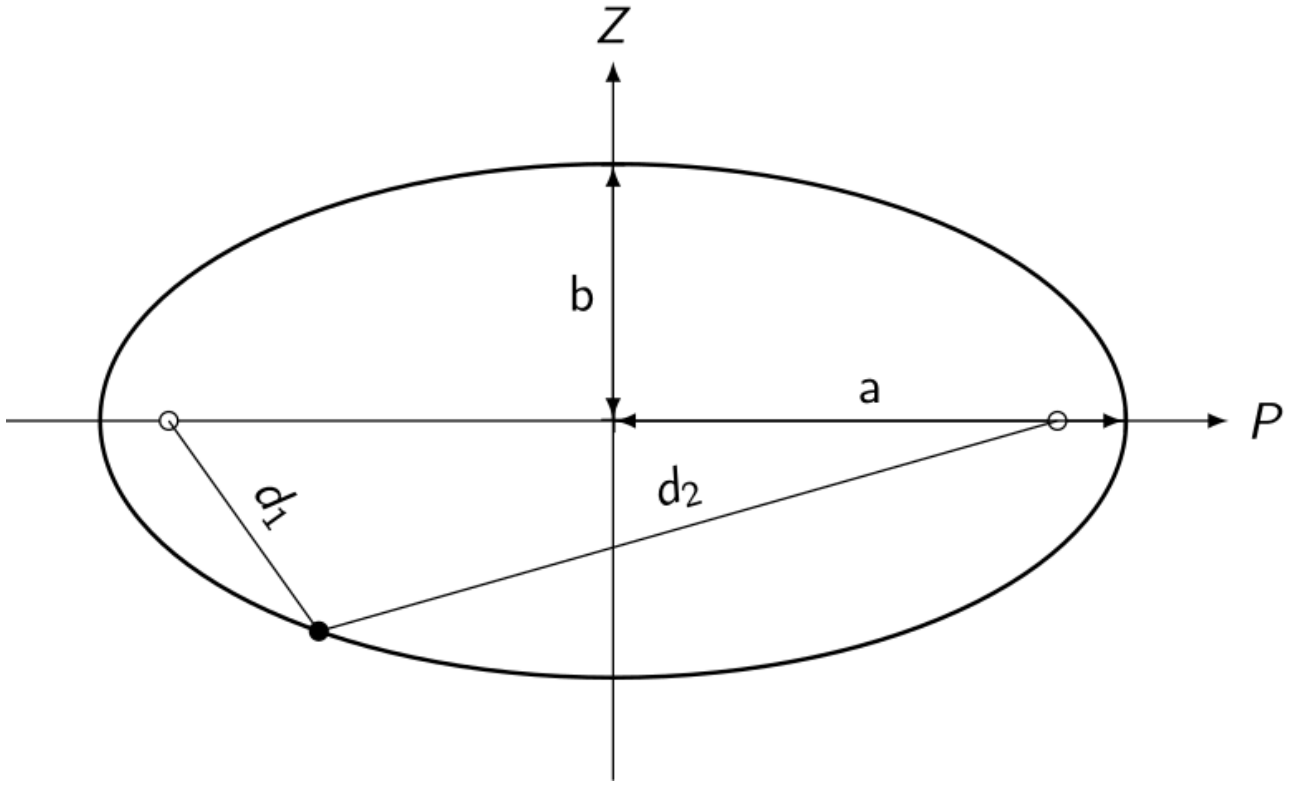


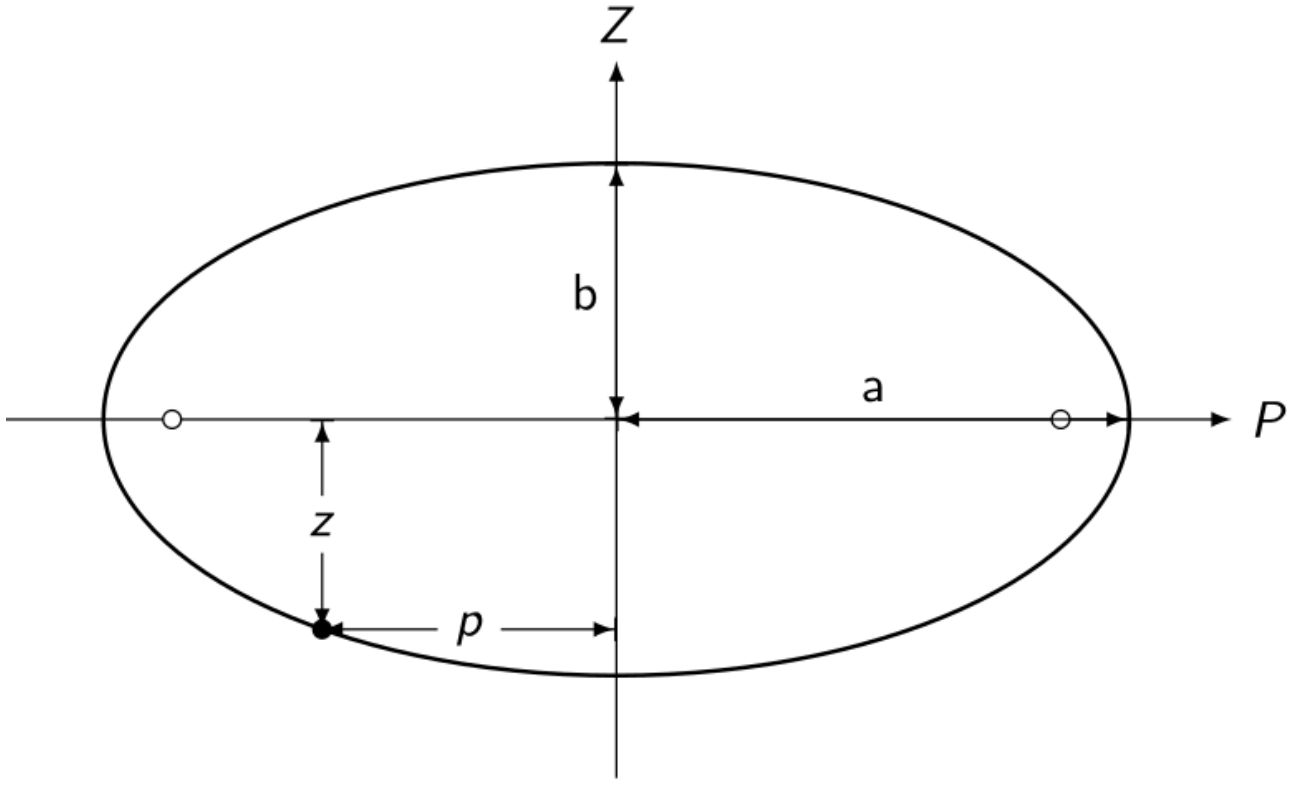
Define 'p-coordinates'

$$p = \sqrt{x^2 + y^2}$$

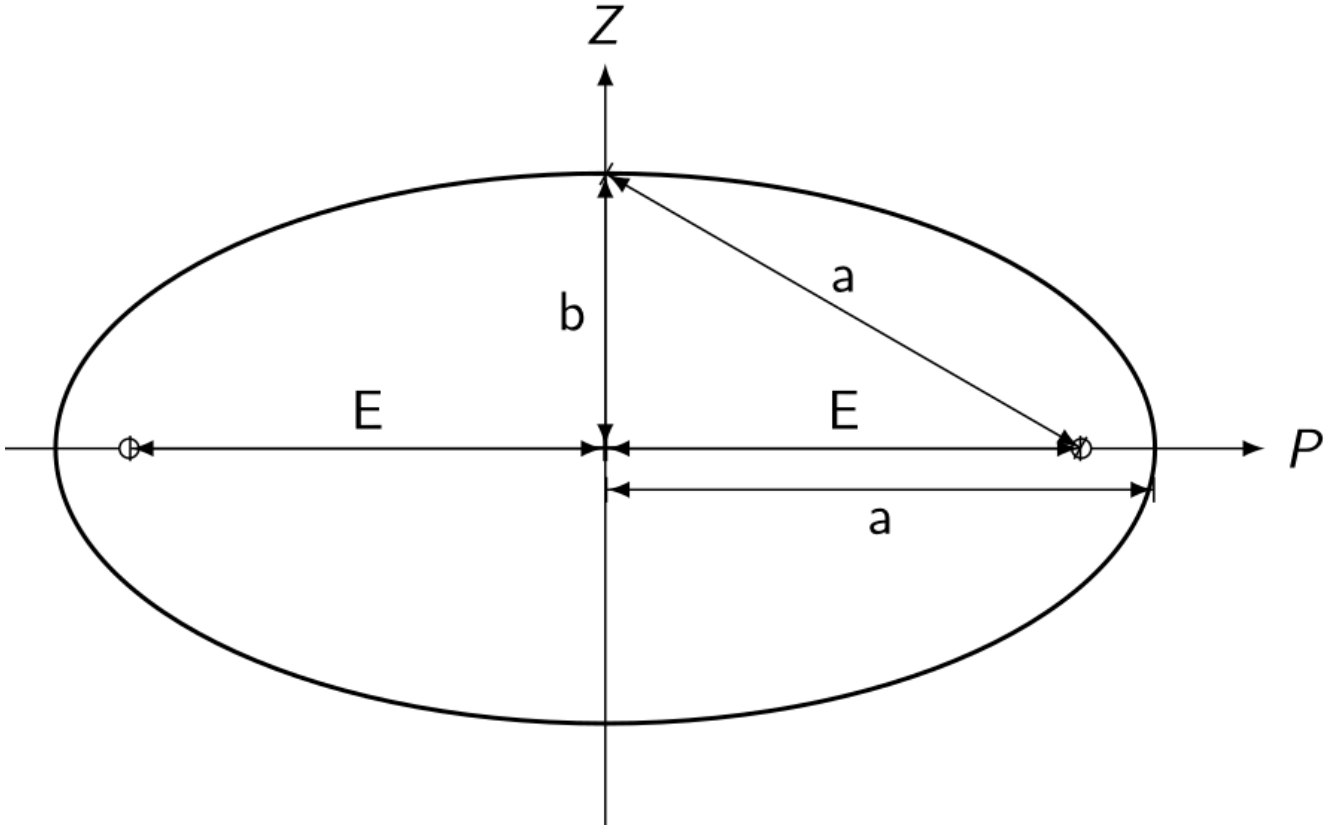
Ellipse

1. $d_1 + d_2 = const = 2a$

- 

2. Equation of Ellipse:
   - $(\frac{p}{a})^2 + (\frac{z}{b})^2 = 1$
   - 

3. Basic laws:

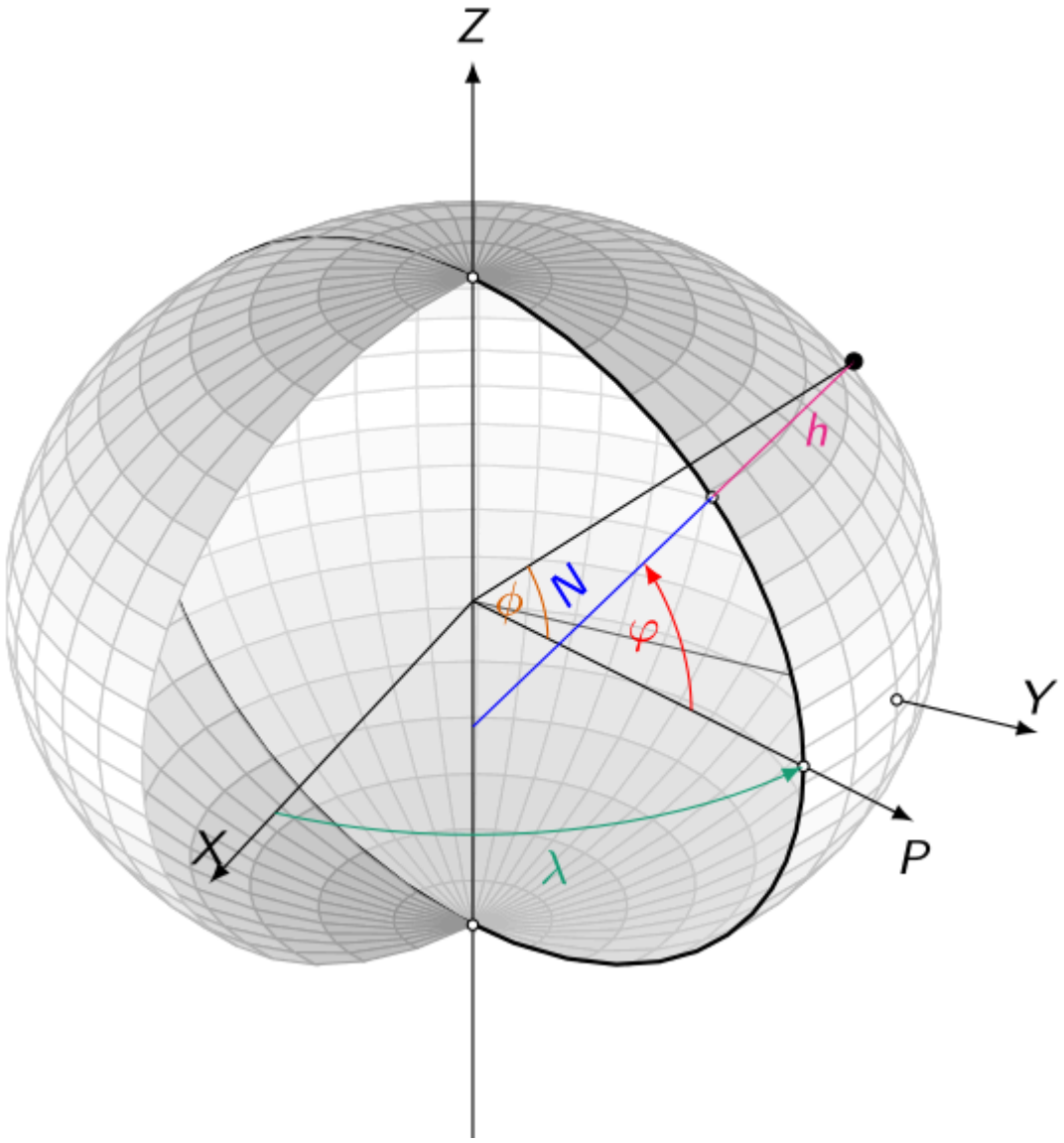


Name	Parameter	Earth
Semi-Major Axis	a	6378.137 km
Semi-Minor Axis	b	6356.752 km
Focal distance	$E = \sqrt{a^2 - b^2} = ea$	521.854 km
First eccentricity	$e = \frac{E}{a}$	0.0818 …
Second eccentricity	$e' = \frac{E}{b}$	0.0821 …
Oblateness	$f = (a - b)/a$	0.003 352 ..

Numerical Precision

- For given ellipsoid: 2 parameters independent
  - alll other derivable
- At Maximum precision: Transformation of coordinates with 0.1 mm accuracy
  - Especially for e, e', f

Parametrisation with ellipsoidal latitude



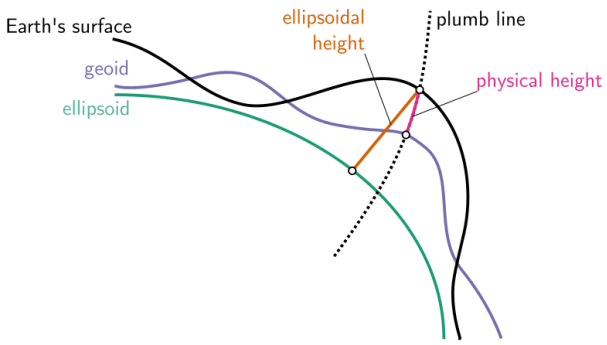
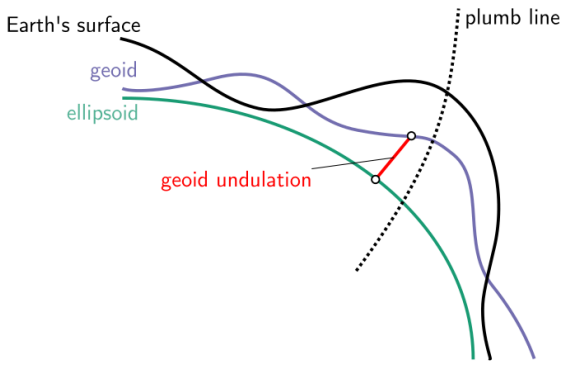
$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2(\varphi)}}$$

$$\boldsymbol{q}(\lambda, \varphi) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (N(\phi) + h) \cos(\lambda) \cos(\varphi) \\ (N(\phi) + h) \sin(\lambda) \cos(\varphi) \\ (\frac{N(\varphi)}{1+e'^2} + h) \sin(\varphi) \end{bmatrix}$$
$$\lambda \in [0^\circ, 360^\circ)$$
$$\varphi \in [-90^\circ, 90^\circ]$$

Ellipsoidal coordinates alone $[\lambda, \varphi, h]$ are useless -> Requirement for:

- Geodetic datum
  - Defines origin, orientation and scale of CS
    - relative to Earth or any other celestial body
- Shape parameters of the ellipsoid
- Location and orientation of ellipsoid

Ellipsoidal and Physical Heights

height	undulation
	

Coordinate Transformations

Ellipsoidal to cartesian

1. Compute transverse radius of curvature:
   $N(\varphi) = \frac{a}{\sqrt{1-e^2 \sin^2(\varphi)}}$
2. Compute x,y,z

$$\boldsymbol{q}(\lambda, \varphi) = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} (N(\phi) + h) \cos(\lambda) \cos(\varphi) \\ (N(\phi) + h) \sin(\lambda) \cos(\varphi) \\ (\frac{N(\varphi)}{1+e'^2} + h) \sin(\varphi) \end{bmatrix}$$
$$\lambda \in [0^\circ, 360^\circ)$$
$$\varphi \in [-90^\circ, 90^\circ]$$

Cartesian to Ellipsoidal Coordinates

Require: $x,\ y,\ z, a,\ b$

$\lambda \leftarrow \arctan\left(\frac{y}{x}\right)$

$e' \leftarrow \sqrt{\frac{a^2-b^2}{b^2}}$

$e \leftarrow \sqrt{\frac{a^2-b^2}{a^2}}$

$p \leftarrow \sqrt{x^2+y^2}$

$h \leftarrow 0$ ▷ first guess for height

$\varphi \leftarrow \arctan\left(\frac{z(1+e'^2)}{p}\right)$ ▷ first guess for latitude

$\varphi' \leftarrow \infty$ ▷ latitude from previous iteration

while $|\varphi' - \varphi| > \epsilon$ **do** ▷ choose threshold according to required precision

$\varphi' \leftarrow \varphi$

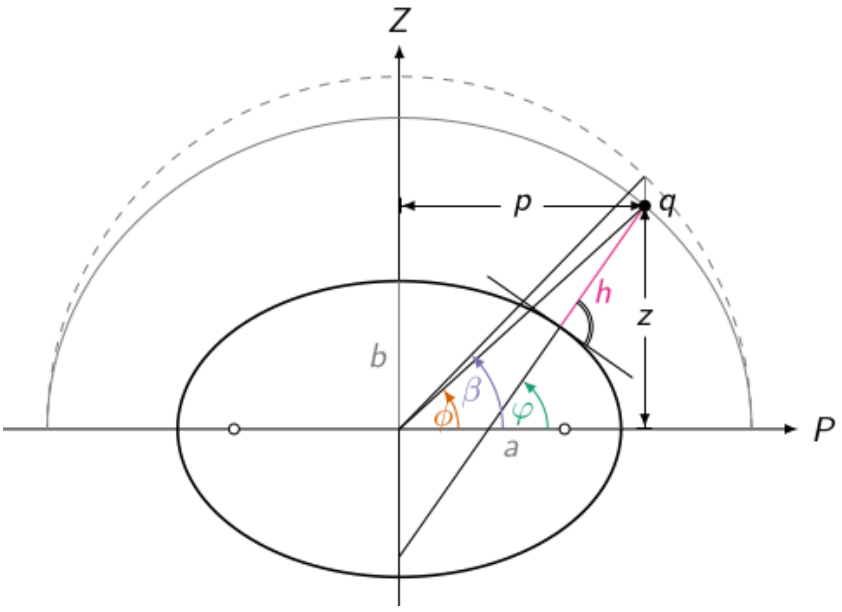
$N \leftarrow \frac{a}{\sqrt{1-e^2\sin^2(\varphi)}}$

$h \leftarrow \frac{p}{\cos(\varphi)} - N$

$\varphi \leftarrow \arctan\left(\frac{z}{p}\frac{N(\varphi)+h}{\frac{N(\varphi)}{1+e'^2}+h}\right)$

end while

Geocentric vs. Ellipsoidal Latitude



$$\phi = \arctan\left(\frac{(1-e^2)N(\varphi)}{N(\varphi)+h}\tan(\varphi)\right)$$

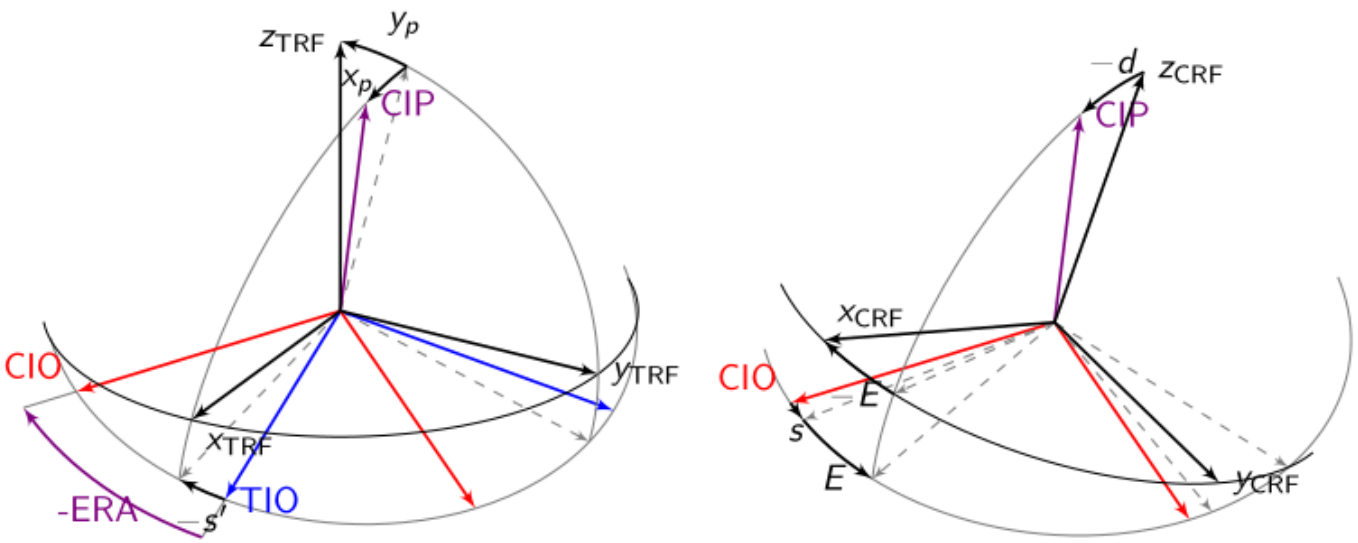
Earths rotation and inertial coordinate system

- Inertial system
  - Force-free objects remain in rest or move with uniform motion

- Also goes for constant rotation **R**

- (Quasi) Inertial reference system
  - Kepler Orbit
  - Practical realisation: GCRF (Geocentric Celestrial Reference Frame)
- Terrestrial reference system (co-rotating with Earth)
  - All terrestrial geodetic markers are located in
  - Reference ellipsoid refers to (z-axis for rotation)

$$\boldsymbol{x}_{\mathrm{GCRS}} = \underbrace{\boldsymbol{R}_3(-E)\boldsymbol{R}_2(-d)\boldsymbol{R}_3(E)\boldsymbol{R}_3(s)}_{\text{precession and nutation } Q(t)} \underbrace{\boldsymbol{R}_3(-ERA)}_{\text{daily rotation } R(t)} \underbrace{\boldsymbol{R}_3(-s')\boldsymbol{R}_2(x_p)\boldsymbol{R}_1(y_p)}_{\text{polar motion } W(t)} \boldsymbol{x}_{\mathrm{ITRS}}$$



For this lecture:

$$\boldsymbol{R} = \boldsymbol{R}_3(\omega_\oplus(t - t_0)$$

Satellite ground tracks

Mean Anomaly M

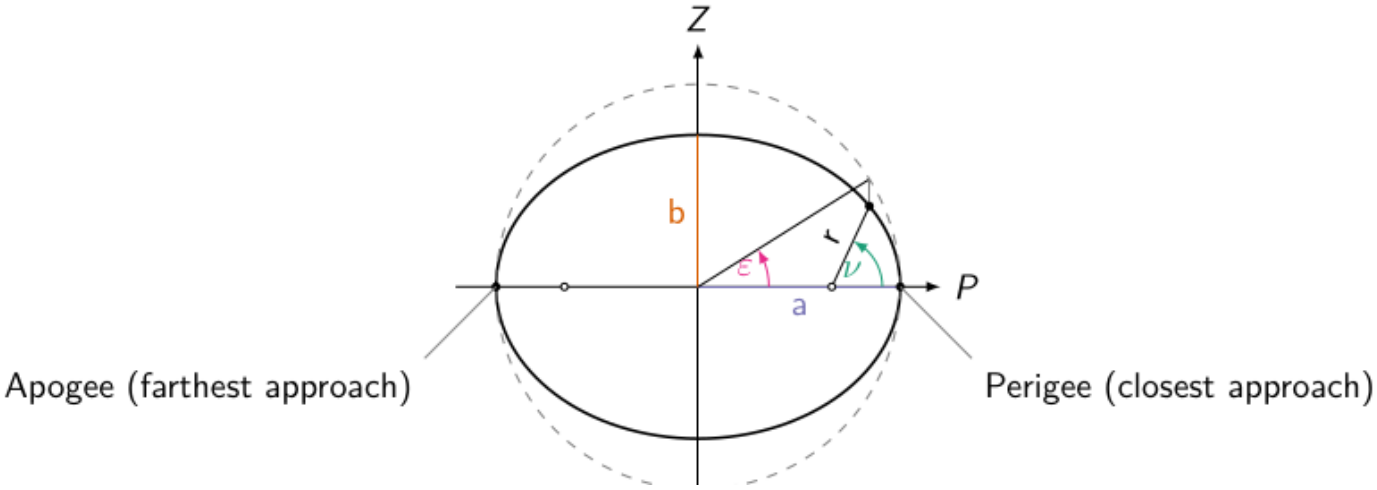
$$M(t) = \tfrac{2\pi}{T}(t - t_0) = \sqrt{\tfrac{GM}{a^3}}(t - t_0)$$

Kepler Equation

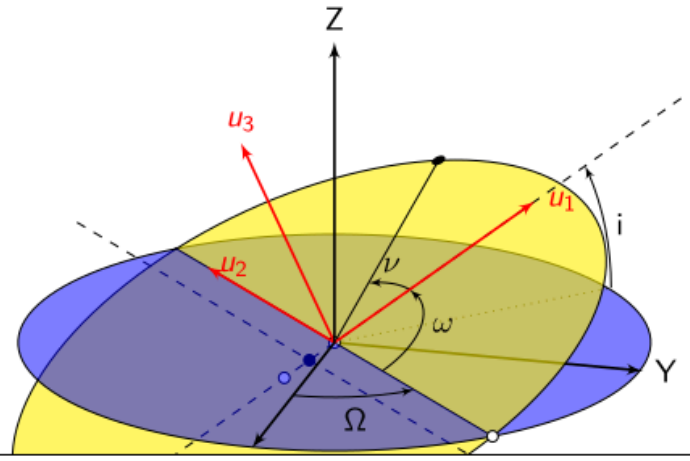
$$M(t) = \underbrace{\varepsilon(t)}_{\text{Eccentric anomaly}} - e\sin(\varepsilon(t))$$

True anomaly ν

$$\cos(\nu(t)) = \tfrac{\cos(\varepsilon(t)) - e}{1 - e\cos(\varepsilon(t))}$$



Transformation: Inertial to orbital coordinate system

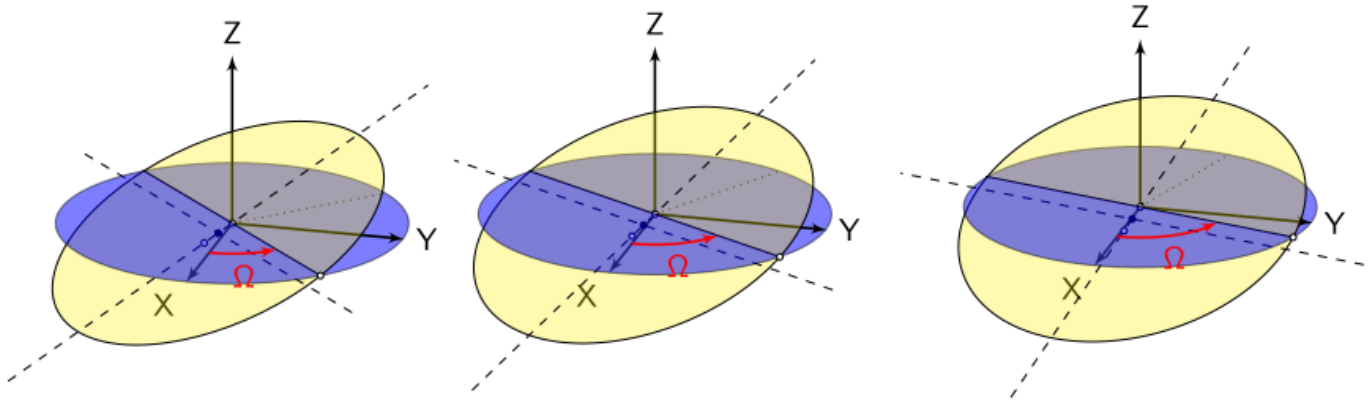


$$\boldsymbol{R} = \boldsymbol{R}_3(\omega)\boldsymbol{R}_1(i)\boldsymbol{R}_3(\Omega)$$

$$\boldsymbol{R} = \begin{bmatrix} \cos(\Omega)\cos(\omega) - \sin(\Omega)\sin(\omega)\cos(i) & -\cos(\Omega)\sin(\omega) - \sin(\Omega)\cos(\omega)\cos(i) & \sin(\Omega)\sin(i) \\ \sin(\Omega)\cos(\omega) + \cos(\Omega)\sin(\omega)\cos(i) & -\sin(\Omega)\sin(\omega) + \cos(\Omega)\cos(\omega)\cos(i) & -\cos(\Omega)\sin(i) \\ \sin(\omega)\sin(i) & \cos(\omega)\sin(i) & \cos(i) \end{bmatrix}$$

Orbital Precession

- Kepler Problem assumes point masses (equivalent to spheres with homogeneous density distribution) -> isotropic gravity field
- But: Earth flattens
- -> Ω changes with time
- "Orbital precession"



Orbitl Precession Rate

Orbital precession rate $\dot{\Omega}$

mean motion

Earth radius

dynamical flattening

$$\dot{\Omega} = -\frac{3R^2 n J_2}{2(1-e^2)^2 a^2}\cos(i)$$

1st eccentricity

semimajor axis

inclination

Orbital precession in Earth fixed frame

$$\dot{\Omega}_e = \dot{\Omega} - \omega_\oplus = -\frac{3R^2 n J_2}{2(1-e^2)^2 a^2}\cos(i) - \omega_\oplus$$

$\omega_\oplus$: Earth rotation rate (do not confuse with argument of perigee)

Approximating Ground Tracks

- For Visualisation:
- Transformation from Keplerian elements to geographical coordinates of sub-satellite point, assuming
  - Circular orbit ($E = e = e' = 0$)
  - Ω, i, ω are constant (point masses)
  - Position along orbit: mean anomaly
- Not accurate at cm-level

$$\phi = \arcsin(\sin(i)\sin(\omega+\nu))$$

$$\lambda = \omega_\oplus(t-t_0) + \Omega + \arctan\left(\frac{\cos(\omega+\nu)\tan(i)}{\sin(\phi)}\right)$$

with $\omega_\oplus = \frac{360^\circ}{\text{sideral day}}$